

Device Physics for Nanoscale CMOS: A Working Revision

KKKC6353 Nanoelectronic Devices · Lecture 1, Part 2 of 3

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How this part works

Common pitfall notes flag explanations from widely used texts that do not survive scrutiny.

1

Carriers and transport

Sections 1 and 2 compress bulk physics. **Read fast.** If any of it is new, go back to Sze or Taur and Ning before Lecture 2.

2

The MOSFET

Section 3 builds the device. **Not optional:** everything in Part 3 is a perturbation on this long-channel MOSFET.

3

Where the picture breaks

Section 4 shows why the classical account is now used past its limit. **This is what makes the module more than an undergraduate course.**

Notation used throughout

Symbol	Meaning	Note
$\mathcal{E}$	Electric field	Never E: that is reserved for energy
$U_T = kT/q$	**Thermal voltage**	25.9 mV at 300 K
V_T	**Threshold voltage**	Not the thermal voltage
ψ_s, ϕ_F	Surface potential, bulk Fermi potential	$\phi_F = U_T \ln(N_A/n_i)$
μ_n, μ_p	**Bulk** mobility	$cm^2/V{\cdot}s$
μ_{eff}	**Inversion-layer** mobility	Not the same quantity (Section 4.3)
τ, λ	Mean free time, mean free path	$\lambda = v_{th}\,\tau$
L	**Physical** gate length	Not the node name

1

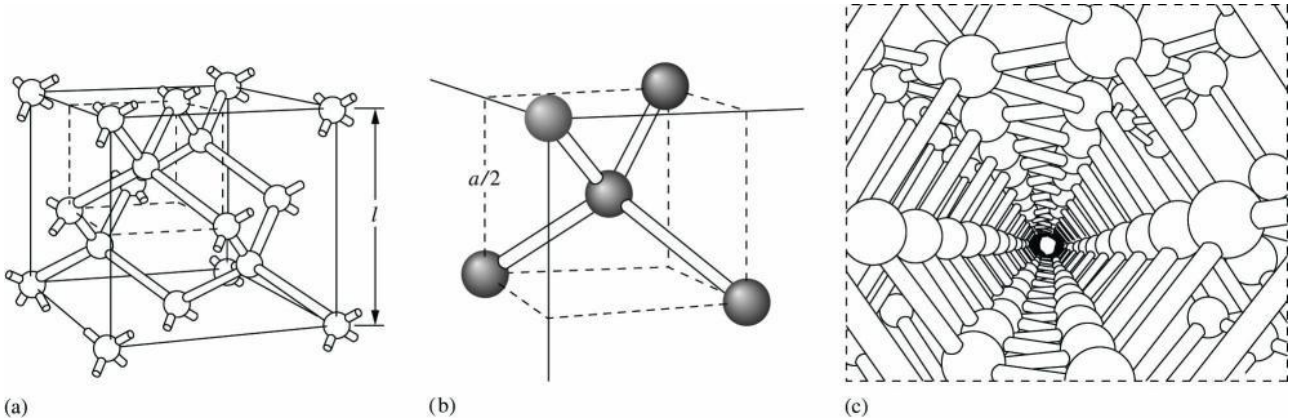
Carriers in silicon

Almost every comparison in this module reduces to effective mass.

Bands and effective mass

- The periodic potential turns the free-electron parabola into **bands**
- Near a band edge the dispersion is parabolic again, so Newton's laws work with an **effective mass** m* in place of m_0
- Effective mass is **inversely proportional to band curvature**: a flat band means a heavy carrier
- Why one material is faster, why strain helps, why holes are slower: all **effective mass**

$$E(k) \approx E_C + \frac{\hbar^2 k^2}{2m^*} \qquad \frac{1}{m^*} = \frac{1}{\hbar^2}\frac{\partial^2 E}{\partial k^2}$$



Silicon: diamond-lattice cell, tetrahedral bonds, view along an axis. Textbook figure © McGraw-Hill; reproduced from S. Levkov, NJIT ECE 271 lecture slides, Ch. 2.

Silicon's two effective masses

	Conduction band	Valence band
Conductivity mass (transport)	$m_c^* \approx 0.26\ m_o$	$\approx 0.39\ m_o$
Density-of-states mass (counting carriers)	$m_d^* \approx 1.08\ m_o$	$\approx 0.81\ m_o$
GaAs, single Γ minimum	$m^* \approx 0.063\ m_o$	

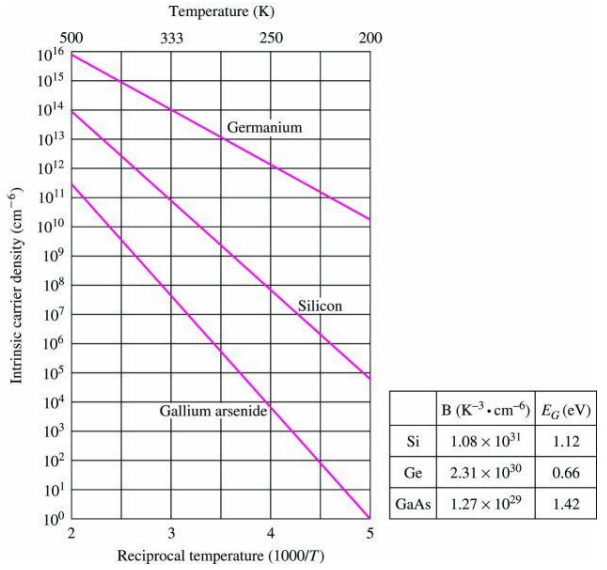
Six conduction-band ellipsoids ($m_l = 0.92$, $m_t = 0.19\ m_o$) need two different averages. GaAs is **four times lighter**: the reason III–V channels interest us.
Pitfall: holes are slower because the valence band is **flatter**, not because of 'bond disruptions'.

Intrinsic carriers

- Boltzmann statistics hold while E_F is a few kT inside the gap
- $N_C \approx 2.8\times10^{19}$, $N_V \approx 1.04\times10^{19}$ cm^{-3} (Si, 300 K)
- Multiplying removes E_F: the **mass-action law**
- **$n_i \approx 9.65\times10^{9}$ cm^{-3}** (Altermatt 2003); older texts quote 1.45×10^{10}

$$n = N_C\, e^{-(E_C - E_F)/kT} \qquad p = N_V\, e^{-(E_F - E_V)/kT}$$

$$np = N_C N_V\, e^{-E_g/kT} = n_i^2$$



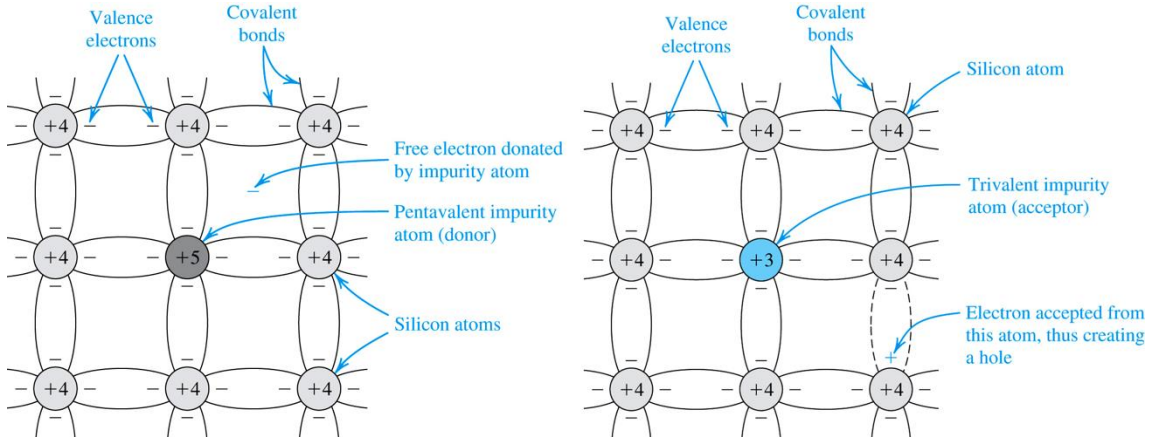
	B ($K^{-3}\cdot cm^{-6}$)	E_G (eV)
Si	1.08×10^{31}	1.12
Ge	2.31×10^{30}	0.66
GaAs	1.27×10^{29}	1.42

n_i against 1/T for Ge, Si, GaAs: the slope is set by the gap. Textbook figure © McGraw-Hill; reproduced from S. Levkov, NJIT ECE 271 lecture slides, Ch. 2.

Doping

	IIIA	IVA	VA	VIA
	5 10.811 **B** Boron	6 12.01115 **C** Carbon	7 14.0067 **N** Nitrogen	8 15.9994 **O** Oxygen
	13 26.9815 **Al** Aluminum	14 28.086 **Si** Silicon	15 30.9738 **P** Phosphorus	16 32.064 **S** Sulfur
IIB				
30 65.37 **Zn** Zinc	31 69.72 **Ga** Gallium	32 72.59 **Ge** Germanium	33 74.922 **As** Arsenic	34 78.96 **Se** Selenium
48 112.40 **Cd** Cadmium	49 114.82 **In** Indium	50 118.69 **Sn** Tin	51 121.75 **Sb** Antimony	52 127.60 **Te** Tellurium
80 200.59 **Hg** Mercury	81 204.37 **Tl** Thallium	82 207.19 **Pb** Lead	83 208.980 **Bi** Bismuth	84 (210) **Po** Polonium

Group III acceptors, group IV semiconductors, group V donors. Textbook figure © McGraw-Hill; reproduced from S. Levkov, NJIT ECE 271 lecture slides, Ch. 2.



A donor's fifth electron is easily freed; an acceptor leaves a hole. Textbook figure © McGraw-Hill; reproduced from S. Levkov, NJIT ECE 271 lecture slides, Ch. 2.

Neutrality with full ionisation: $\mathbf{n \approx N_D - N_A}$ and $\mathbf{p \approx n_i^2/n}$ for n-type material with $N_D - N_A \gg n_i$.

Where Boltzmann statistics stop working

Degenerate source and drain

Modern S/D doping exceeds **10^{20} cm^{-3}**, pushing E_F into the conduction band. Boltzmann fails; the full **Fermi–Dirac** integral is needed.

Why it matters now

$np = n_i^2$ no longer holds there, and **bandgap narrowing of ~100 meV** sets in. Neither touches the channel, but both affect **source resistance and injection**, which Section 4 shows now limits drive current.

2

Transport

Keep the mean free path in view: it decides whether any of this applies.

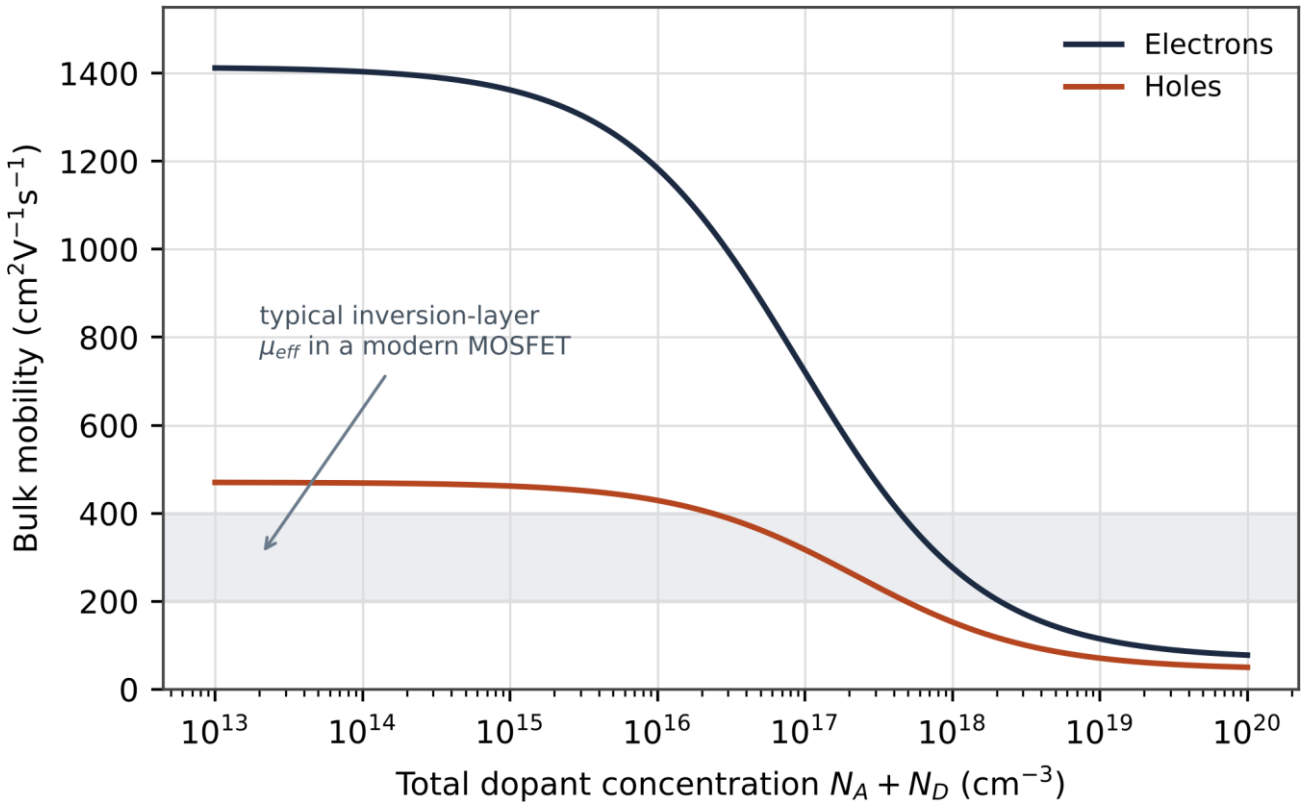
Scattering and mobility

- Carriers move at $\mathbf{v_{th} \approx 2.3\times10^7\ cm/s}$ in random directions; no net current
- Collisions with phonons, dopants and interfaces every **τ**; mean free path $\boldsymbol{\lambda = v_{th}\,\tau}$
- A field adds a steady **drift** on top of the thermal motion
- **Two levers only:** rare collisions (large τ) and light carriers (small m*)
- **Pitfall:** μ = qτ/2m* assumes every free flight lasts exactly τ. Free times are random and exponentially distributed, which gives **qτ/m***

$$v_d = -\mu_n \mathcal{E} \qquad \mu = \frac{q\tau}{m^*}$$

Mobility depends on doping and temperature

- **Low doping:** phonon scattering dominates; μ_n up to ~1414 cm²/V·s
  - μ falls as T rises (the lattice vibrates harder)
- **High doping:** ionised-impurity scattering dominates
  - μ rises as T rises (faster carriers are deflected less)

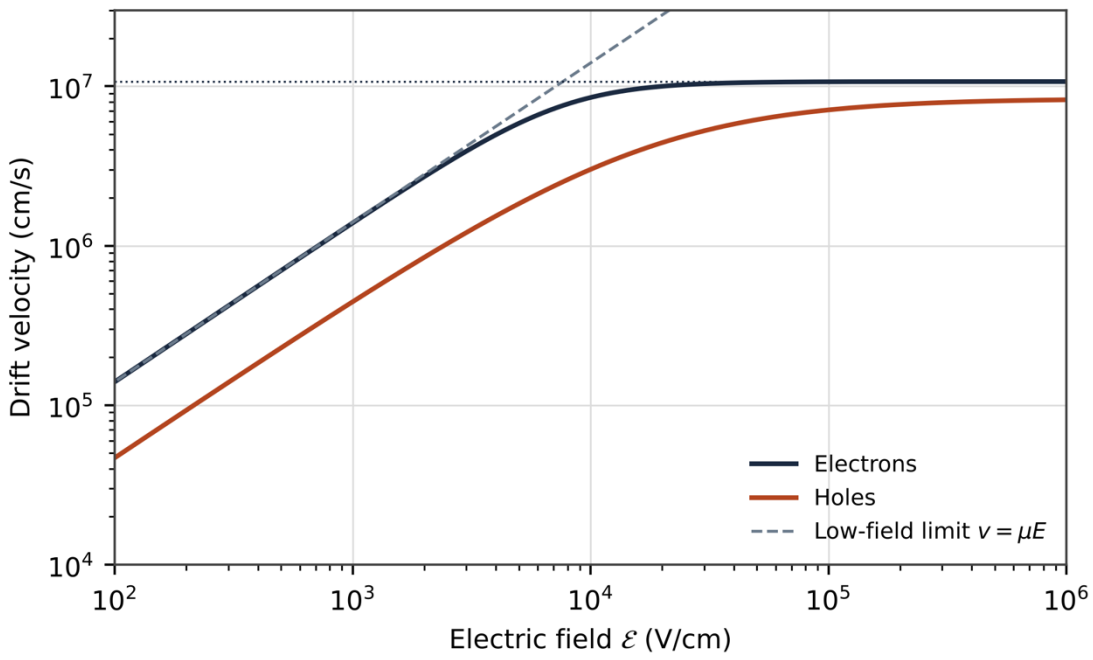


Bulk Si mobility at 300 K, Caughey–Thomas model.

High fields: velocity saturation

- Once a carrier can emit an **optical phonon**, energy is removed as fast as the field supplies it
- Velocity levels off at $\mathbf{v_{sat} \approx 10^7}$ **cm/s**
- **A modern transistor never works in the low-field regime**
- So the square law overestimates drive current, and Part 3 treats saturation as a short-channel effect

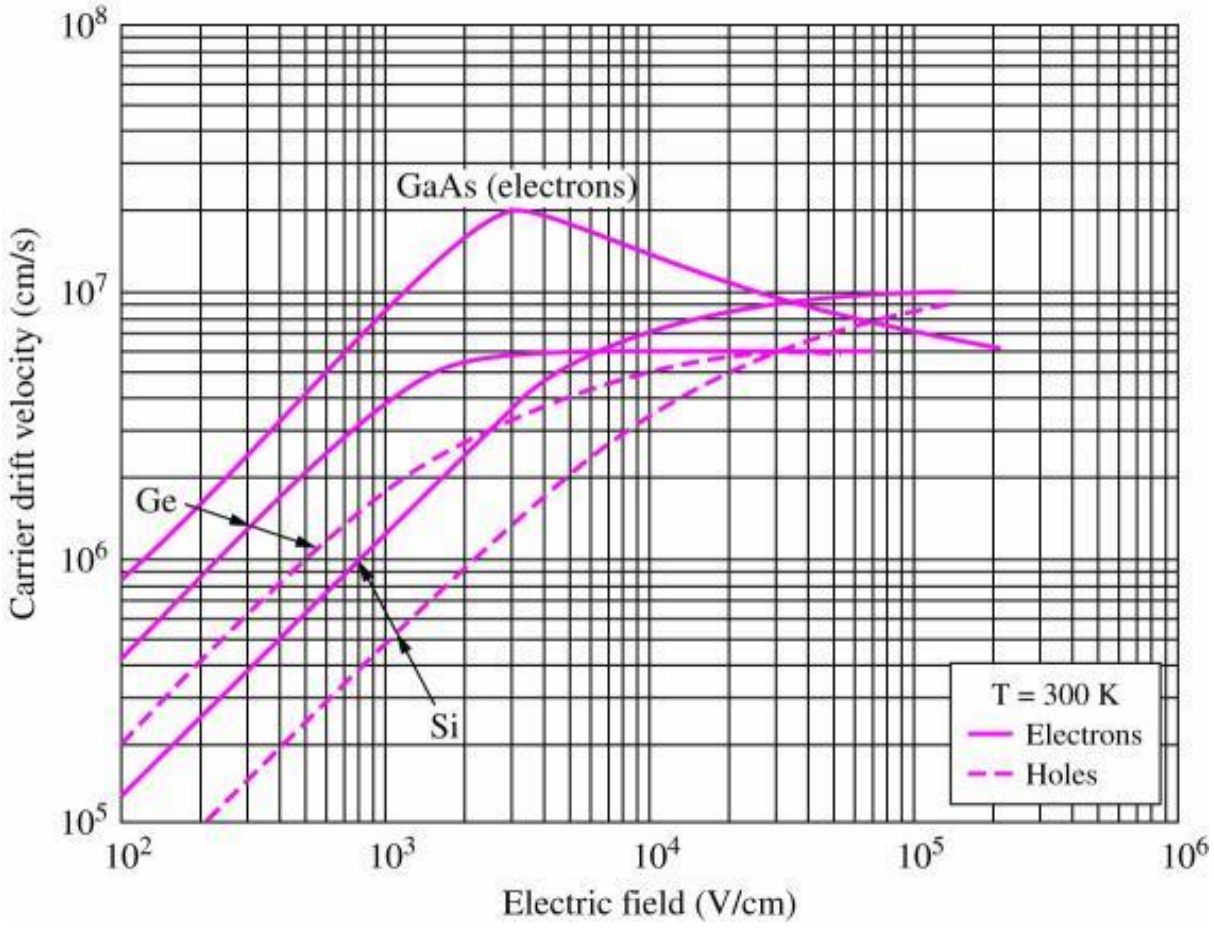
$$v = \frac{\mu \mathcal{E}}{\left[1 + (\mu \mathcal{E}/v_{sat})^{\beta}\right]^{1/\beta}}$$



Electrons and holes in Si at 300 K; dashed: $v = \mu \mathcal{E}$.

Velocity against field in three semiconductors

- Electrons are faster than holes in every material: lighter m*
- **GaAs electrons overshoot** and then fall back as they transfer to heavier valleys
- The basis of the Gunn effect
- At high field the materials converge near **10^7 cm/s**



Measured drift velocity, Ge, Si and GaAs, 300 K. Textbook figure © McGraw-Hill; reproduced from S. Levkov, NJIT ECE 271 lecture slides, Ch. 2.

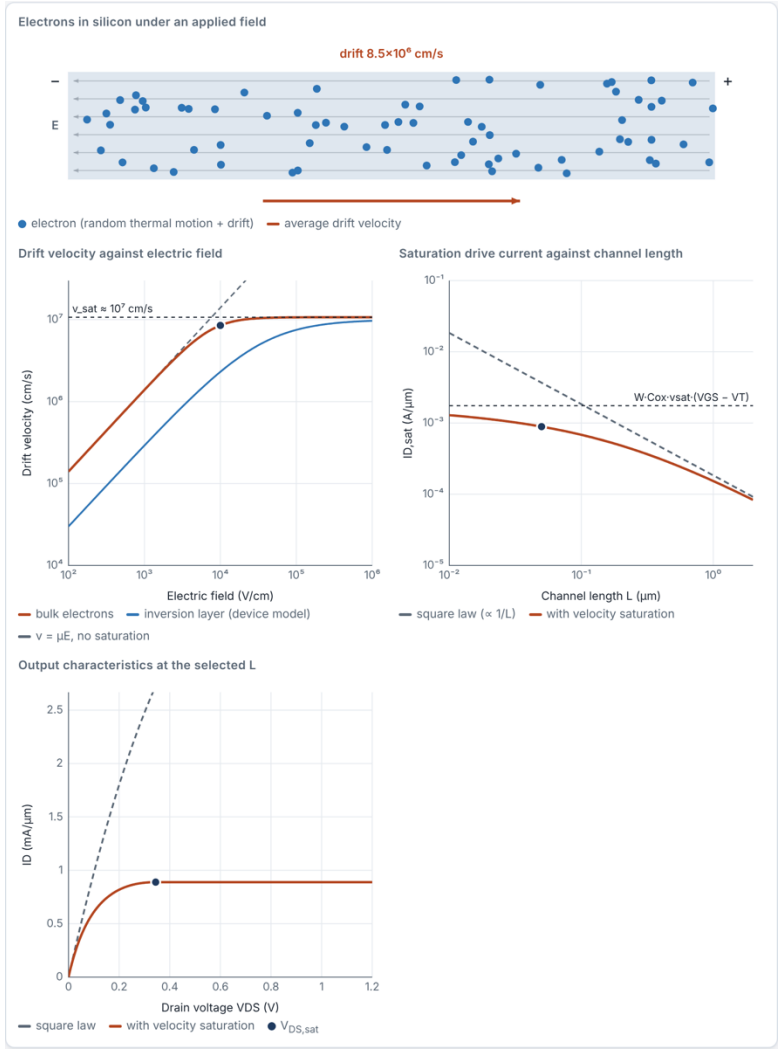
Animation: carrier velocity saturation

Interactive animation 7

Carrier velocity saturation

faizverse.github.io/nanoelectronic-devices/L1/velocity-saturation.html

- Field at 10^3 V/cm: electrons jitter but barely drift
- Raise to 10^5 V/cm: drift catches up with thermal speed, then stops growing
- Shrink L to 20 nm: saturation current stops following 1/L

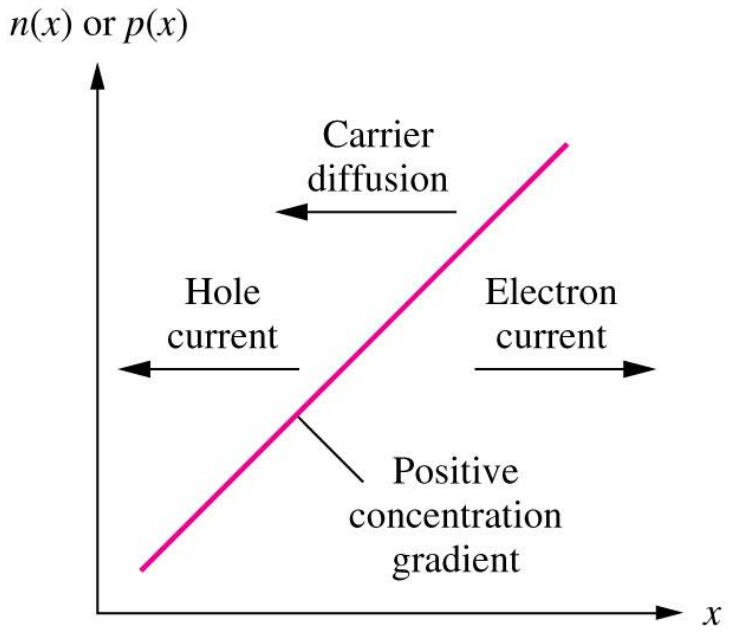


Diffusion and the Einstein relation

- A concentration gradient drives a flux even with no field (Fick's law)
- Drift and diffusion come from the **same collisions**, so D and μ are linked
- **U_T = 25.9 mV** at 300 K (25 mV is the value at 290 K)
- Holds for **non-degenerate** material only

$$\frac{D_n}{\mu_n} = \frac{D_p}{\mu_p} = \frac{kT}{q} = U_T = 25.9 \text{ mV}$$

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Electron diffusion current flows opposite to the electron flux. Textbook figure © McGraw-Hill; reproduced from S. Levkov, NJIT ECE 271 lecture slides, Ch. 2.

Drift-diffusion, and what it assumes

- **1. Many collisions:** $\lambda \ll L$, so a drift velocity is meaningful
- **2. Local equilibrium:** potential varies slowly over λ
- **3. Classical particles:** no quantised energy levels
- **4. Non-degenerate statistics** for the Einstein relation
- **Every one is violated somewhere in a modern transistor.** Hold that thought until Section 4

$$J_n = q\mu_n n\mathcal{E} + qD_n\frac{dn}{dx} \qquad J_p = q\mu_p p\mathcal{E} - qD_p\frac{dp}{dx}$$

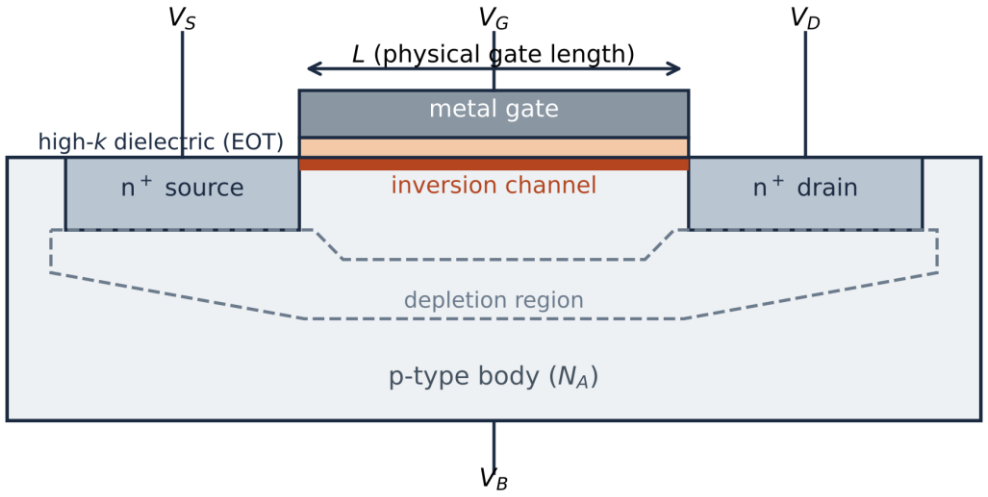
The workhorse of classical device physics and of every TCAD default.

3

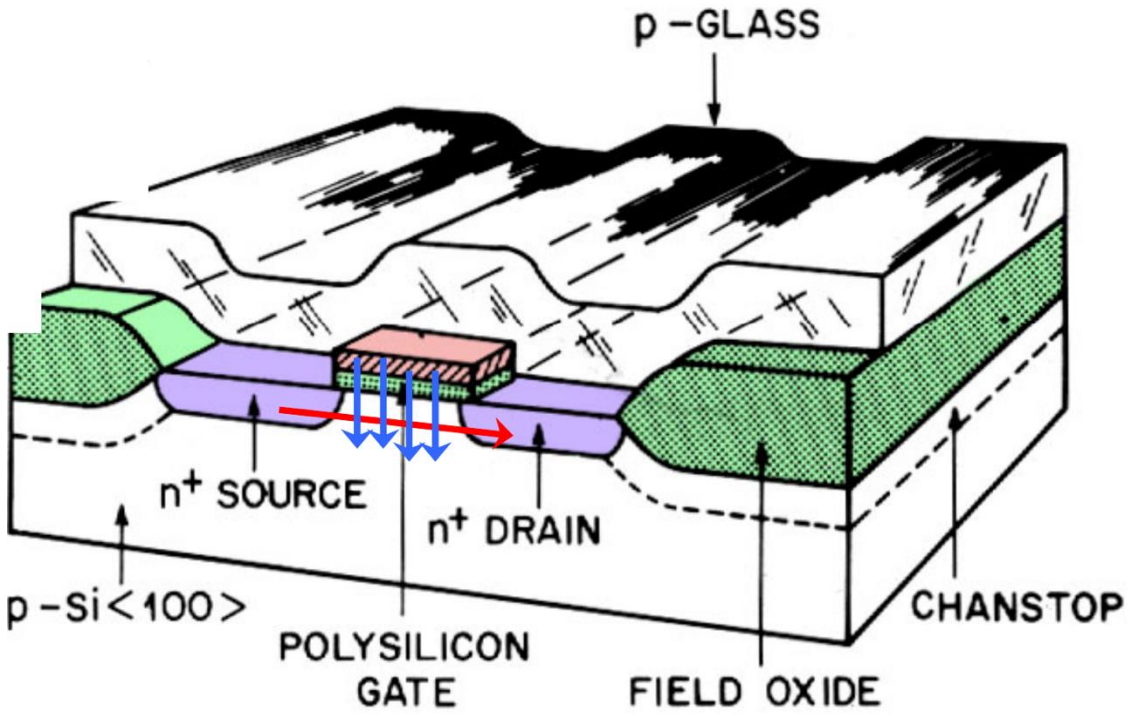
The MOSFET

Everything in Part 3 is a perturbation on this device.

The device and its terminology



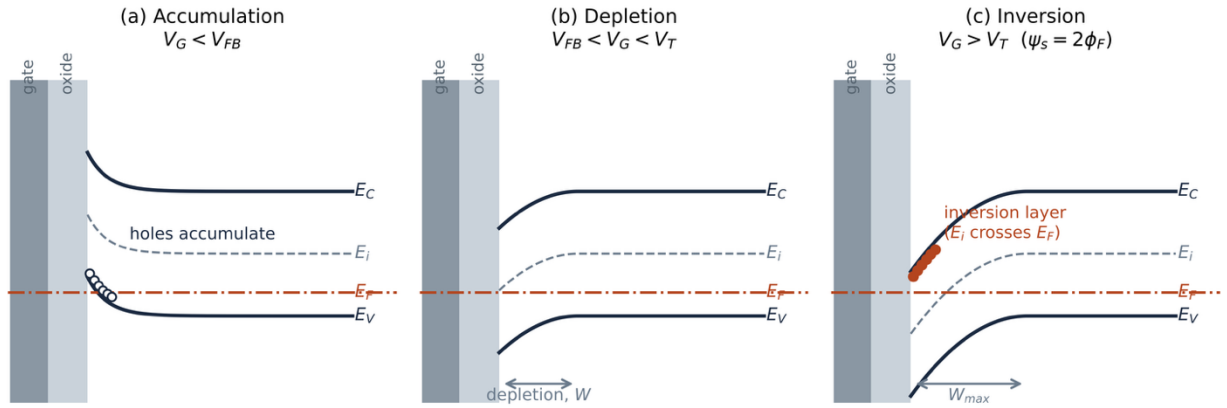
n-channel MOSFET; L is the physical gate length, not the node name.



Planar MOSFET of the LOCOS era. Original source not stated; reproduced from W. A. Doolittle, Georgia Tech lecture slides (2011).

The MOS capacitor: three regimes

- **Accumulation** ($V_G < V_{FB}$): holes pulled to the surface; bands bend up; no channel
- **Depletion** ($V_{FB} < V_G < V_T$): holes pushed away, leaving fixed acceptor ions
- **Inversion** ($V_G > V_T$): the surface turns n-type; a sheet of electrons forms the **channel**
- Strong inversion at $\boldsymbol{\psi_s = 2\phi_F}$, a convention: charge below it falls off **exponentially** (Section 3.5)



$$W = \sqrt{\frac{2\varepsilon_{Si}\psi_s}{qN_A}} \qquad \psi_s = 2\phi_F, \quad \phi_F = U_T \ln\frac{N_A}{n_i}$$

Threshold voltage and gate capacitance

- V_T rises with N_A (through ϕ_F and the square root)
- V_T falls as C_{ox} rises (electrically thinner dielectric)
- **The gate is assumed to support all the depletion charge.** In a short channel it does not: Part 3
- Modern devices set V_T by **metal work function**, not channel doping

$$V_T = V_{FB} + 2\phi_F + \frac{\sqrt{2q\varepsilon_{Si}N_A(2\phi_F)}}{C_{ox}}$$

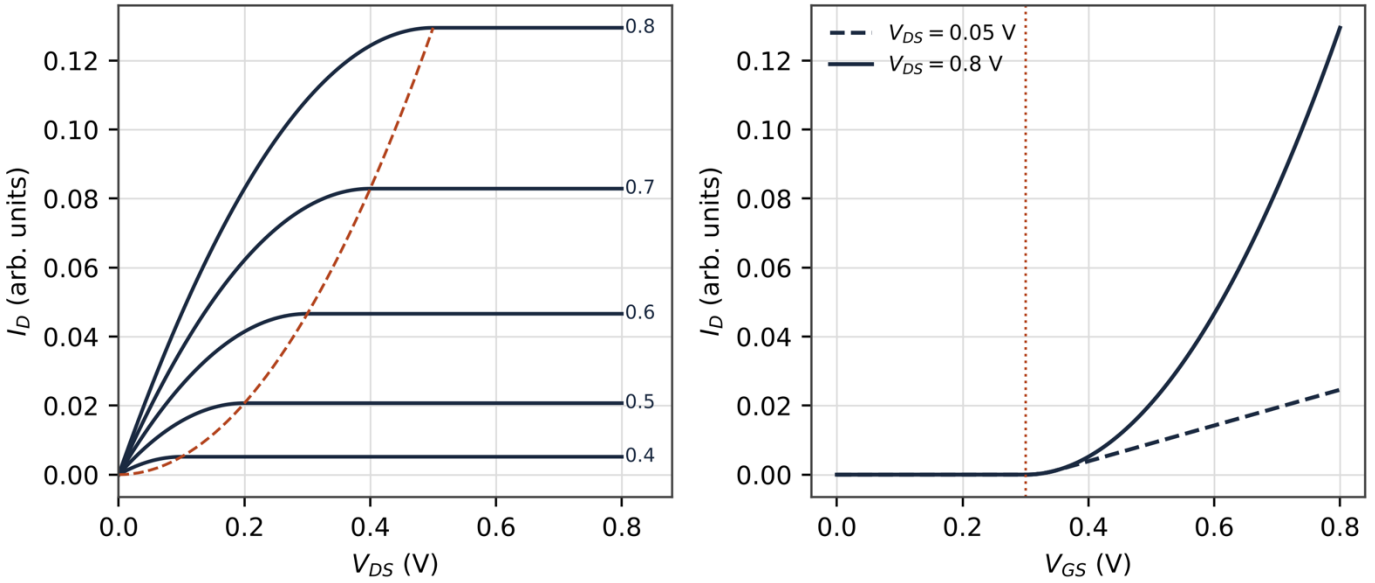
$$C_{ox} = \frac{\varepsilon_0 k}{t_{phys}} = \frac{\varepsilon_0 k_{SiO_2}}{\mathrm{EOT}}$$

EOT = 1 nm gives C_{ox} = 3.45 µF/cm² = 34.5 fF/µm².

Long-channel current–voltage: the square law

- **Gradual channel:** charge set by the local vertical bias, $Q_{inv} = -C_{ox}(V_{GS} - V_T - V)$
- Saturation at $\mathbf{V_{DS} = V_{GS} - V_T}$: pinch-off at the drain
- **Warning:** assumes constant mobility. With velocity saturation, $I_D \approx W C_{ox} v_{sat} (V_{GS} - V_T)$: **linear**, independent of L
- A scaffold for understanding, not a design tool

$$I_D = \mu_{eff} C_{ox} \frac{W}{L}\left[(V_{GS} - V_T)V_{DS} - \frac{V_{DS}^2}{2}\right] \qquad I_{D,\,sat} = \frac{\mu_{eff} C_{ox}}{2}\frac{W}{L}(V_{GS} - V_T)^2$$



EOT 1 nm, μ_{eff} = 300 cm²/V·s, V_T = 0.3 V.

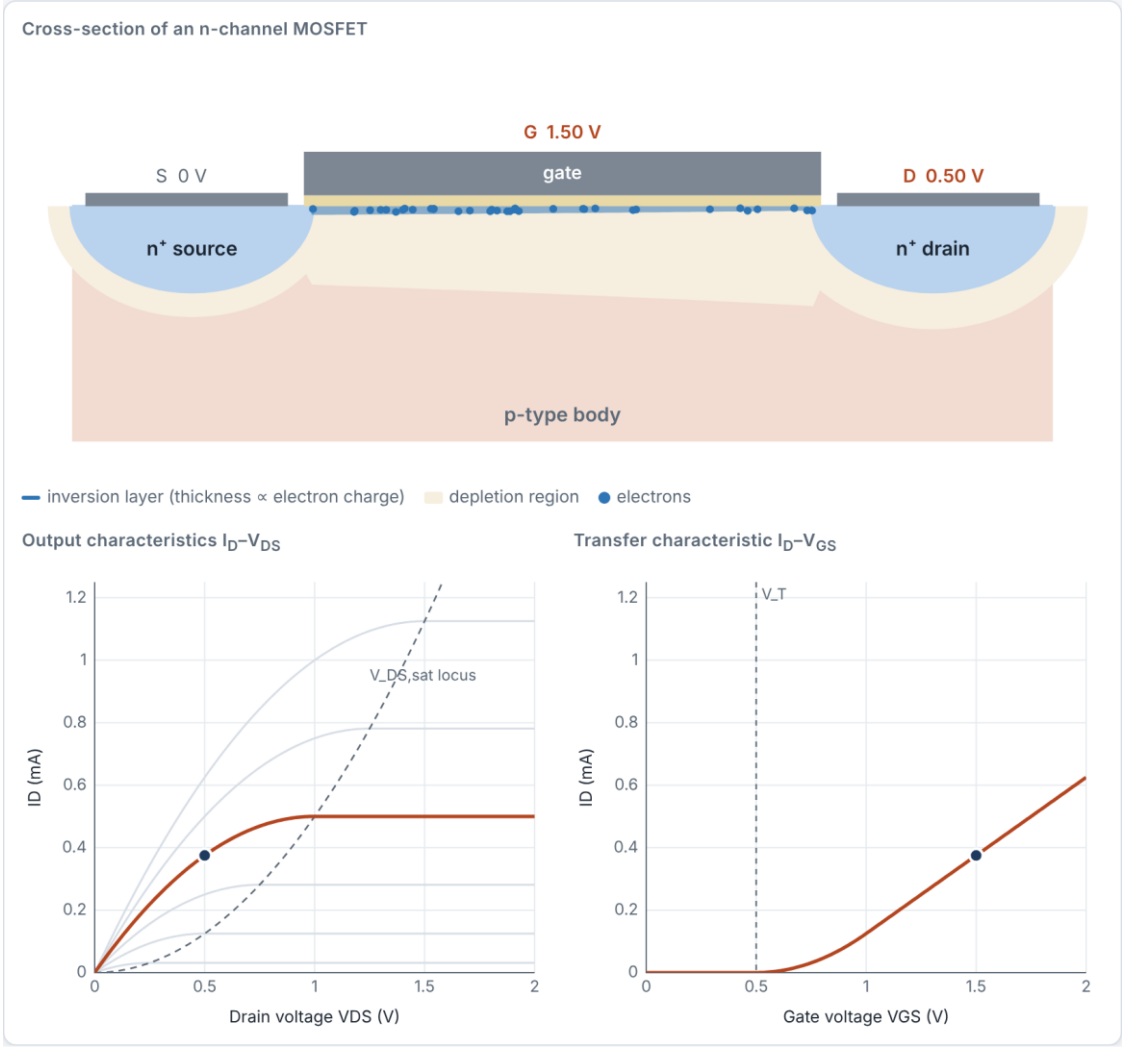
Animation: MOSFET basics

Interactive animation 1

MOSFET basics

faizverse.github.io/nanoelectronic-devices/L1/mosfet-basics.html

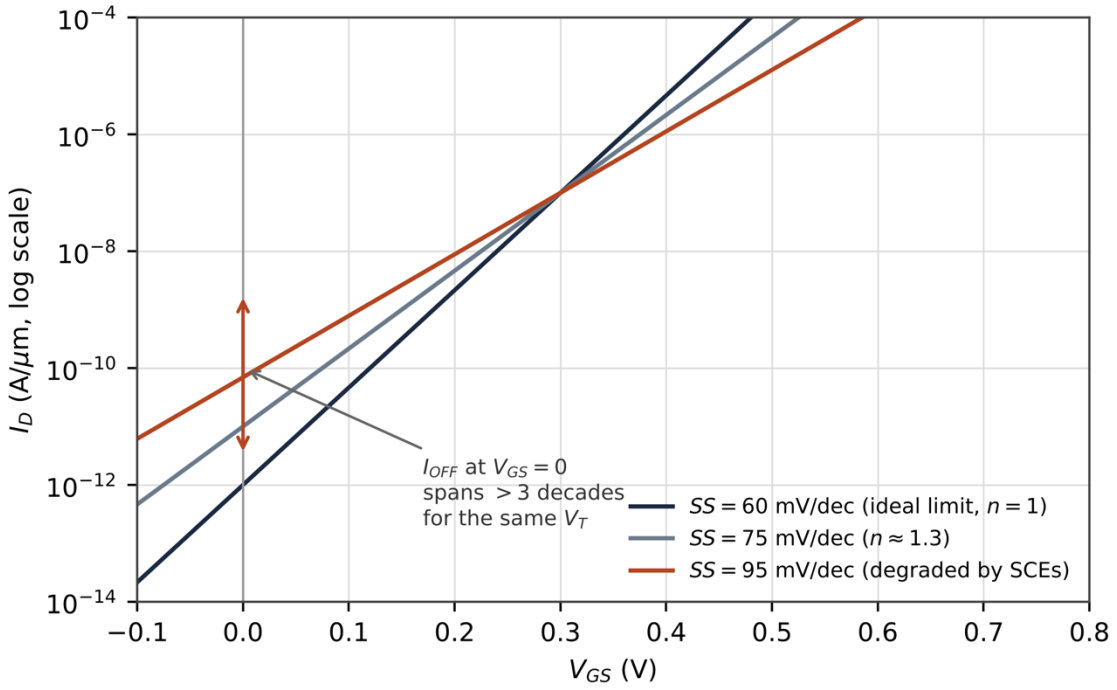
- V_{DS} = 0.1 V, raise V_{GS}: nothing flows until the inversion layer appears
- Fix V_{GS} = 1.5 V, raise V_{DS}: the layer tapers and pinches off at 1.0 V
- Watch electrons speed up where the channel thins: continuity



Subthreshold conduction: the 60 mV/decade limit

- **The most important result in Lecture 1**
- Below threshold the channel is not off: current falls **exponentially**, carried by diffusion
- $n = 1 + C_d/C_{ox}$ is a **capacitive divider**: the gate shares control with the depletion region
- Perfect control ($n \to 1$) still gives **59.5 mV/dec**: a **thermodynamic** limit from the Boltzmann tail

$$SS = \frac{dV_{GS}}{d\log_{10} I_D} = \left(1 + \frac{C_d}{C_{ox}}\right)\frac{kT}{q}\ln 10 \;\geq\; 59.5 \text{ mV/dec}$$



Same V_T, three swings: the off-current at $V_{GS} = 0$ differs by orders of magnitude.

Why this ended Dennard scaling

59.5 mV

per decade at 300 K: the floor, set by **temperature alone**

10×

more leakage for every **60–100 mV** taken off V_T, on every transistor

C_d/C_{ox}

the **only** term engineering can touch: thin-body SOI, FinFET, GAA

$I_{OFF} \propto 10^{(-V_T/SS)}$. Constant-field scaling demanded lower V_T every node; by the early 2000s, 10^8 devices × leakage rivalled switching power. **Steep-slope devices** (tunnel FETs, negative-capacitance FETs) try to escape the Boltzmann tail altogether (Ionescu and Riel, Nature 2011).

Body effect

- Reverse body bias widens the depletion region: the gate supports more charge, so **V_T rises**
- Heavily used in circuit design
- It largely **disappears** in fully depleted thin-body and GAA devices: no quasi-neutral body left to bias
- A little-discussed consequence of the FinFET transition

$$V_T(V_{SB}) = V_{T0} + \gamma\left(\sqrt{2\phi_F + V_{SB}} - \sqrt{2\phi_F}\right)$$

$$\gamma = \frac{\sqrt{2q\varepsilon_{Si}N_A}}{C_{ox}}$$

γ: the body-effect coefficient. A thin, fully depleted body has no N_A to bias.

4

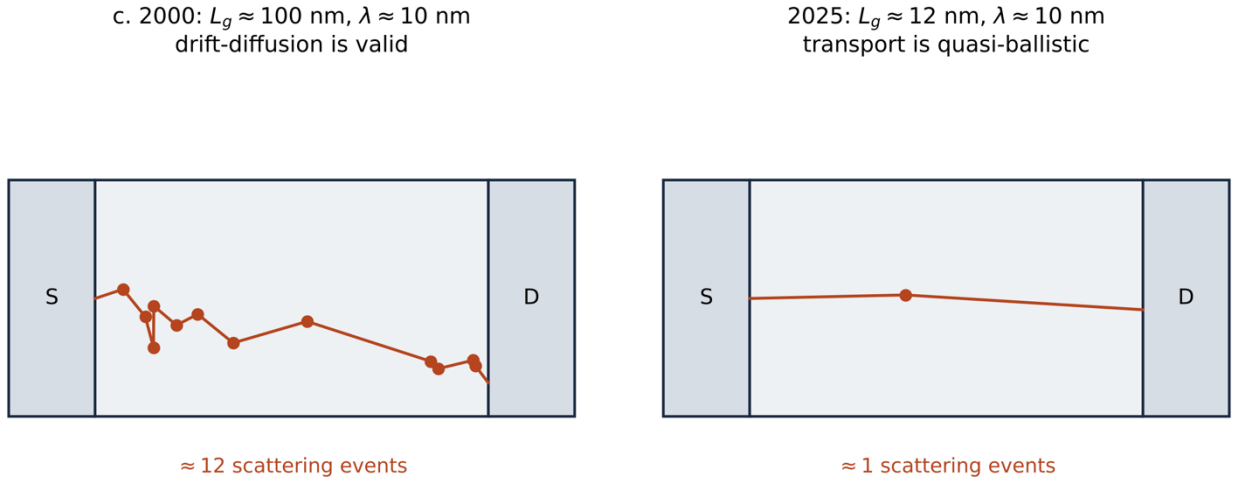
Where this picture breaks

A competent account of a 1 µm transistor, applied at 15 nm.

λ ≈ L: transport is quasi-ballistic

- Inversion-layer electrons: **λ ≈ 10 nm**
- Gate length in production: **12–20 nm**, so **L/λ ≈ 1–2**
- Mobility and drift velocity are no longer clean averages; drift-diffusion is **extrapolated**
- **Pitfall:** comparing λ with a 100 nm gate and concluding 'many collisions' was right in 2000, not now

$$\tau = \frac{\mu_{eff} m_n^*}{q} \approx 44 \text{ fs} \qquad \lambda = v_{th}\tau \approx 10 \text{ nm}$$

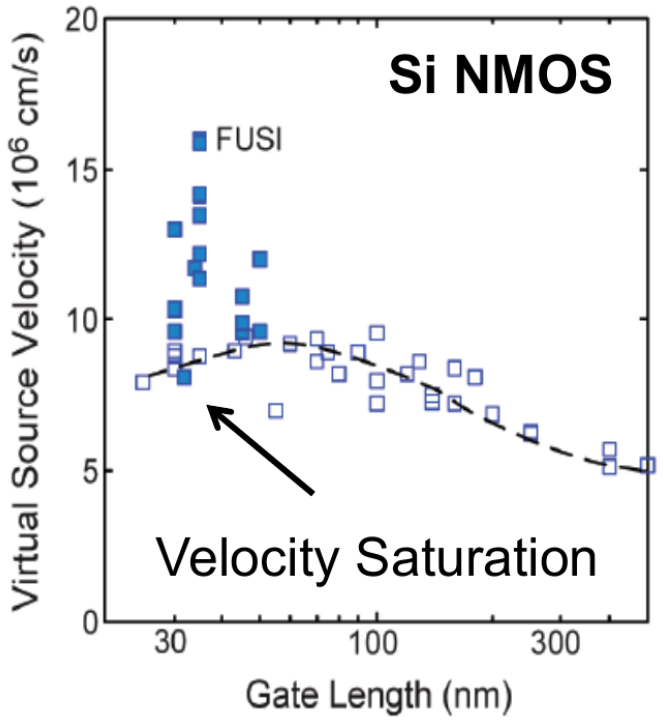


Same mean free path; a 100 nm gate (c. 2000) and a 15 nm gate (today).

When injection, not transport, sets the current

- Ballistic limit (Natori, Lundstrom): current set by the **source injecting carriers** over the barrier
- $v_{inj} \sim 10^7$ cm/s, **independent of L**
- Shortening L no longer raises drive current
- **Source engineering** (resistance, injection velocity) becomes the lever
- Mobility survives as a **backscattering** figure of merit: why strain still works

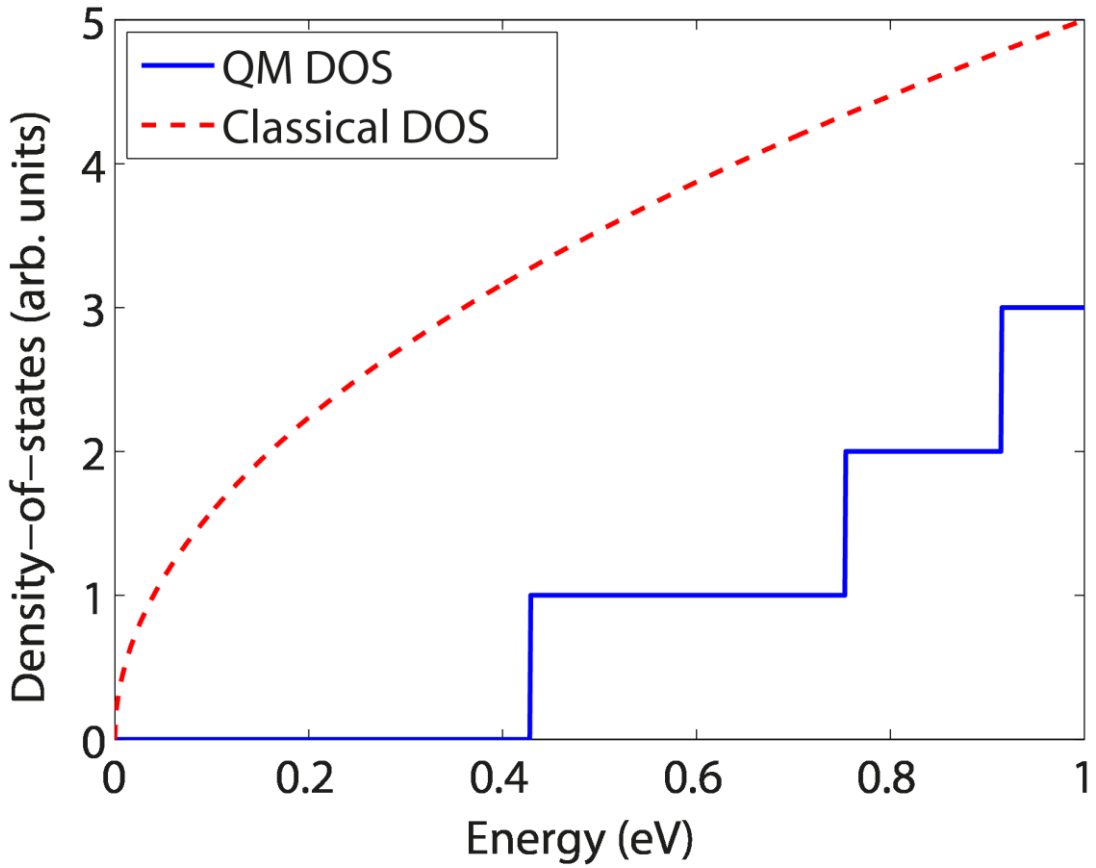
$$I_{D,\,\text{ballistic}} = W\,Q_{inv}\,v_{inj}$$



Measured injection velocity against gate length. A. Khakifirooz and D. A. Antoniadis, IEEE TED 55, 1391 (2008); reproduced from ETH Zürich slides.

Inversion-layer quantisation

- The channel is a potential well **1–2 nm** wide: a two-dimensional electron gas
- **Lowest state above the band edge:** V_T rises by tens of mV, up to ~100 mV
- **Charge centroid ~1 nm** from the interface: adds 0.3–0.4 nm to the effective EOT, a **30–40% penalty** at EOT 1 nm, irreducible
- **Density of states** becomes a staircase
- Real devices need Schrödinger–Poisson (TCAD)



Density of states in a thin channel (solid) against bulk (dashed). Reproduced from ETH Zürich slides.

Effective mobility is not bulk mobility

Bulk μ_n reaches **1414** $cm^2/V \cdot s$; in a modern inversion layer $\boldsymbol{\mu_{eff} \approx 200–400}$. Three causes:

Surface roughness

Dominant at high vertical field: carriers pressed against an interface that is not atomically flat.

Remote Coulomb and phonon

Charges and polar vibrations in the **high-k** dielectric: one cost of hafnium oxides.

Confined phonons

Stronger phonon scattering in the two-dimensional system. Data collapse onto a **universal mobility curve**.

Pitfall

Not 'the wavefunction extends into the oxide'. The evanescent tail carries no current.

What survives at L ≈ 15 nm

Concept	Status
Band structure, effective mass	**Valid.** More necessary than before
Doping, mass action, ϕ_F	**Valid** in the channel; fails in degenerate S/D
Electrostatics, V_T, C_{ox}	**Valid**, with quantum corrections to V_T and EOT
Subthreshold swing, 60 mV/dec	**Valid and binding.** Thermodynamic
Mobility as a transport coefficient	**Questionable.** Better read as a backscattering figure of merit
Drift-diffusion transport	**Outside its domain.** Usable with calibration
Square-law I_D	**Pedagogical only.** Velocity saturation dominates

The electrostatics survive; the transport does not. That is why every architectural change of the last fifteen years (thin body, FinFET, GAA) is an electrostatic fix.

What to carry forward

1 $\mu = q\tau/m^*$

Light carriers and rare collisions. Two levers, nothing else.

2 $U_T = 25.9\ \text{mV}$

kT/q at 300 K, and V_T is a **different** quantity.

3 $\psi_s = 2\varphi_F$

Inversion begins there; V_T is what it costs to get there.

4 $SS \geq 59.5\ \text{mV/dec}$

Thermodynamic. The only handle is C_d/C_{ox}.

5 $\lambda \approx 10\ \text{nm} \approx L$

Transport is quasi-ballistic; drift-diffusion is extrapolated.

6 **Next: Part 3**

So far only the gate controls the channel. When it is short, **the drain competes**.

Figure credits

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Textbook figures © McGraw-Hill, reproduced from S. Levkov, NJIT ECE 271 lecture slides (Ch. 2): crystal, n_i, periodic table, doping, velocity–field, diffusion. MOSFET perspective drawing reproduced from W. A. Doolittle, Georgia Tech lecture slides (2011). Injection velocity: A. Khakifirooz and D. A. Antoniadis, IEEE TED 55, 1391 (2008), and density of states, both reproduced from ETH Zürich lecture slides. Copyright remains with the original owners.

Made for this course

Mobility, velocity–field, MOSFET cross-section, MOS band diagrams, long-channel and subthreshold characteristics, and scattering diagrams were generated for KKKC6353. Equations typeset for these slides. Animation screenshots from faizverse.github.io/nanoelectronic-devices.